

CS-SOFTWARE ASSIGNMENT-3

MVC Design Pattern and Its Variants

The Model-View-Controller (MVC) design pattern divides software into three interconnected components: the Model, the View, and the Controller.

Model: Manages data and business logic.

View: Displays data (the user interface).

Controller: Handles user input and updates the Model and View accordingly.

Variants:

1. Model-View-Presenter (MVP): Presenter replaces the Controller. It retrieves data from the Model and formats it for display in the View.

2. Model-View-ViewModel (MVVM): ViewModel acts as a bridge between the View and Model, typically used in data-binding scenarios.

Appropriateness:

- Use MVC when separation of concerns is needed with minimal data-binding.
- Use MVP for testability in GUI applications.
- Use MVVM for rich UI applications with data binding (e.g., WPF, Angular).